Canary: A Pre-Registered Single-Signal Novelty Study on SEC 10-K MD&A Text

Arsenii Chan
CSC 44800, Artificial Intelligence
Spring 2026 · City College of New York
Final Project Report
Repository: github.com/ArseniiChan/canary · Live demo: canary-psi.vercel.app · Frozen spec tag: validation-spec-frozen

Abstract

Can a computer reading the words in an annual report tell which companies turned out to be committing accounting fraud? This project tests one specific answer. I treat each Form 10-K filing's Management's Discussion and Analysis section (Item 7) as a sequence of sentences, embed every sentence with all-MiniLM-L6-v2, train a small autoencoder (384 → 128 → 32 → 128 → 384) on clean peer filings, and score each filing by mean per-sentence reconstruction error. Training is leave-one-cohort-out and time-controlled: the model only sees clean peer filings from other cohorts dated on or before each fraud's filing date. The held-out evaluation set is six historical accounting frauds: Enron (FY2000), WorldCom (FY2001), Tyco (FY2001), HealthSouth (FY2001), Valeant (FY2014), and Lehman Brothers (FY2007). The analysis specification, including exact accession numbers, was committed and git-tagged validation-spec-frozen before any validation script ran.

The headline has three constraints. The pre-registered design was infeasible for Enron: under the strict LOCO + time-controlled rule no eligible training data exists, because Enron's filing date precedes every peer in every other cohort. Enron is excluded from the primary results, and the exclusion is itself a finding about pre-registration, not fraud detection. Of the five remaining cohorts, only Lehman ranks in the top three (rank 3/13, sentence-level Mann-Whitney $p \approx 1.3 \times 10^{-9}$, rank-biserial effect +0.12); the other four (HealthSouth 7/12, Tyco 9/13, Valeant 8/13,

WorldCom 12/13) rank no better than the random-baseline expected rank ($(N+1)/2 \approx 7$). Aggregate hit-rates fall at or below random expectation. A post-hoc TF-IDF + truncated-SVD trivial baseline matches or exceeds the autoencoder on every cohort and produces a stronger Lehman signal ($p \approx 5 \times 10^{-22}$, rank 1/13). On this dataset, the 2020s neural model shows no advantage over a 1990s latent-semantic-analysis baseline. The contribution is limited: a frozen pre-registered evaluation that produced a null result, plus an explicitly post-hoc baseline check that further constrained its interpretation.

1. Introduction and Driving Question

The Management's Discussion and Analysis (MD&A) section of a 10-K is the longest stretch of free-form prose in a public company's annual report. The MD&A is Item 7 in the standard SEC form, where management explains the year in their own words. A literature in accounting and finance argues that companies that later turn out to have been committing fraud leave detectable linguistic fingerprints in their pre-discovery filings. The natural question is whether unsupervised novelty detection, treated as sentence embeddings plus a reconstruction-error model trained on clean filings, can recover that fingerprint without ever seeing a fraud label.

I tested one such answer. The driving question is:

Does an unsupervised autoencoder trained on industry-and-year-matched clean 10-K MD&A text assign elevated reconstruction error to known historical fraud filings, under strictly out-of-sample evaluation?

The one-line answer this report defends: at $N = 6$ frauds, with a frozen pre-registered specification and strict leave-one-cohort-out plus time-controlled training, the answer is no for four of five evaluable cohorts; the one that ranks high (Lehman) is matched or exceeded by a

1990s-era TF-IDF baseline; and the sixth (Enron) cannot be evaluated under the pre-registered rules at all.

Section 2 states the methodological contract that governs every downstream choice. Section 3 describes the data construction. Section 4 covers MD&A extraction. Section 5 specifies the single signal. Section 6 covers the statistical defense, including an explicitly post-hoc TF-IDF baseline check added after observing the primary autoencoder result. Section 7 reports per-fraud results. Section 8 collects conclusions, limitations, and future work.

Canary is not a fraud detector. The project is a comparative novelty study with N = 6 positive examples, evaluated as a held-out probe set under a contract written down and immutable before any number was looked at. The contract prevented tuning after seeing the results; the numbers reported here are what that one-pass evaluation produced. The components map to the CSC 44800 lecture sequence: autoencoder (AI 27, 35); MiniLM embeddings (AI 33, Meetings 22-23); PyTorch implementation (AI 28, 31, 10); post-hoc TF-IDF baseline (AI 12).

Table 0. Mapping of project components to CSC 44800 lectures.

Component	Course coverage
MiniLM sentence embeddings (the input to every model)	AI 33 Attention and transformers; AI 35 Encoders and decoders; Meetings 22–23 (LLMs, Apr 30 / May 5)
Custom autoencoder, 384 → 128 → 32 → 128 → 384 ReLU, MSE, Adam	AI 27 Neural networks; AI 35 Encoders and decoders; AI 28 Tensors; AI 31 Computational graphs; AI 10 Optimizers
Post-hoc TF-IDF + truncated-SVD baseline	AI 12 Introduction to machine learning
Mann-Whitney U + bootstrap + null permutation	Adjacent: Russell & Norvig, general framing of empirical evaluation

2. The Methodological Contract

The plan went through multiple structured pre-registration review rounds before any code ran. The output is analysis_spec.md, committed to the repository and tagged validation-spec-frozen. Once tagged, the file is immutable: scripts/06_validate.py runs once against the frozen spec and produces a single set of numbers reported as-is. The full forensic record is on the repository: commit 98d2585, tag validation-spec-frozen, datestamp 2026-05-01 13:42:42 EDT.

Three rules govern the analysis. The first is leave-one-cohort-out: for each held-out fraud cohort C, the autoencoder is trained on clean peer filings from cohorts other than C, and the fraud filing in C and its peers are never in the training set. The second is time-controlled training: within that LOCO training set, only filings dated on or before each fraud's filing date are eligible, so the autoencoder's training corpus excludes later filings. The frozen MiniLM encoder is pretrained on general web text and is not time-controlled, which is named as a separate contamination caveat in Section 8. The third is single-pass, no tuning: architecture, hyperparameters, sentence cap, seeds, and the test set were all committed before validation, and no tuning of any of these choices took place after observing per-fraud rank numbers.

A peer is admitted to a cohort iff three rules hold: the CIK (Central Index Key, the SEC's per-company identifier) is not on an editable AAER (Accounting and Auditing Enforcement Releases) deny-list (sourced from public knowledge of SEC enforcement releases); the same CIK has filed no Form 10-K/A within five years post-filing; the CIK or company name is not on an editable class-action deny-list. Both deny-lists are committed at data/processed/aaer_denylist.txt and data/processed/classaction_denylist.txt.

What this project deliberately does not do: call itself a fraud detector; train any supervised classifier on fraud labels; use Benford's Law (Benford's Law applies to financial-statement line items, not narrative prose, so it was removed from the spec before any code ran); include foreign-issuer 20-Fs; compose multiple signals into a "Canary score"; claim that any fraud would have been "flagged N months early"; or modify the primary configuration after observing primary results.

3. Data

The held-out evaluation set is six 10-K filings, all pre-discovery (Table 1). Each fraud's accession was verified against a documented public revelation date before being committed to the spec.

Table 1. Held-out fraud filings. CIKs are in the repo manifest at data/processed/fraud_manifest.json.

#	Company	FY	Filed	SIC	Days pre-disc.
1	Enron Corp.	2000	2001-04-02	6200	197
2	WorldCom Inc.	2001	2002-03-13	4813	104
3	Tyco International Ltd.	2001	2001-12-28	3585	157
4	HealthSouth Corp.	2001	2002-03-27	8060	357
5	Valeant Pharmaceuticals	2014	2015-02-25	2834	236
6	Lehman Brothers Holdings	2007	2008-01-29	6211	230

Each cohort is constructed as same SIC (Standard Industrial Classification) 2-digit and same-fiscal-year peers, screened against the operational clean rule. Final cohort sizes: Enron 12, HealthSouth 11, Tyco 12, Valeant 12, Lehman 12, WorldCom 12 clean peers. None required SIC-1-digit fallback.

Two limitations of the data construction matter for interpretation. First, AAER and class-action screening were applied via editable deny-lists rather than systematic database lookups. A peer that committed undisclosed fraud could appear in a cohort, contaminating the peer pool and weakening interpretability of the result. Second, EDGAR reports each company's currently-reported SIC, which may differ from its at-time-of-filing SIC. I use what EDGAR currently reports and apply it symmetrically across fraud and peer filings. The rule above is the cleanest available, though not a perfect control for the firm's sector at the time of filing.

4. MD&A Extraction

The parser at engine/parsing.py handles three filing formats (SGML-era plain-text, early HTML, modern HTML) through a prioritized candidate list of (Item 7 start, Item 7A end) boundary pairs, with fallbacks for filings whose MD&A header lacks a literal "Item 7" prefix (Tyco-style incorporation by reference). Bodies under 3,000 or over 400,000 characters are rejected.

I committed in advance to a hard 80% extraction-success gate before proceeding. The parser hits 73 of 78 filings (93.6%) across all six cohorts. All six fraud filings parsed successfully (100%); the five failures are pre-2001-vintage peer filings with missing primary-document attachments or missing Item 7A end-anchors. No manual extraction fixes were applied. Every parse is the parser's first attempt, so manual correction cannot bias the fraud-vs-peer comparison. Per-cohort detail is in reports/parsing_qa.md.

5. The Single Signal

Each filing's MD&A body is sentence-tokenized using a regex-based splitter that protects common abbreviations. Sentences shorter than 20 characters or composed mostly of digits are dropped. Each remaining sentence is embedded with sentence-transformers/all-MiniLM-L6-v2 (384 dimensions); embeddings are cached on disk per (filing, model) for deterministic reproducibility.

I implement a small symmetric autoencoder in PyTorch at engine/autoencoder.py: 384 → 128 → 32 → 128 → 384 with ReLU activations, MSE reconstruction loss, Adam optimizer at learning rate 1e-3, batch size 16, maximum 200 epochs with early stopping (patience 20) on a 20% validation split. The autoencoder follows the dimensionality-reduction-by-reconstruction paradigm of Hinton & Salakhutdinov (2006): a bottleneck network trained to reconstruct its input forces a lossy compressed representation, and reconstruction error on out-of-distribution inputs is higher than on in-distribution inputs. The 32-dimensional bottleneck is small relative to the 384-dim input by design. Random seeds for numpy, torch, and Python random are all 42.

For each held-out fraud cohort C, the training corpus is constructed by replaying the spec's rule: every sentence from every clean peer filing in cohorts other than C, restricted to filings dated on or before C's fraud filing date. A per-filing sentence cap of 100 prevents long-MD&A pseudoreplication during training. The five trained autoencoders had training-set sizes of 1,477 (WCOM), 861 (TYC), 2,283 (HRC), 4,327 (VRX), and 3,367 (LEH) sentences. Section 8 examines whether the factor-of-five spread between Tyco and Valeant explains per-cohort rank ordering; it does not.

At inference, each filing's MD&A is embedded the same way; the autoencoder's per-sentence reconstruction error is MSE between input and output for that sentence; the filing's score is the mean per-sentence reconstruction error. Per-filing scores for fraud and clean peers in each of the five LOCO cohorts are written to data/results/scores.csv and feed Section 6.

Two design choices that did not work were caught in the build before the spec was frozen and are documented for transparency. First, an early version of this pipeline used Benford's Law on MD&A text as an auxiliary signal. Benford's Law applies to financial-statement line items, not narrative prose, so the method was incoherent on its face; I removed it from the spec before any code ran. Second, the original peer-matching rule used SIC-1-digit codes; preliminary cohort assembly produced peer pools that were too heterogeneous (SIC-1 mixes consumer durables with industrial machinery, for example), so I tightened the rule to SIC-2-digit before freezing the spec. The autoencoder bottleneck dimension itself was frozen at 32 a priori without a sensitivity sweep; that is named as a limitation in Section 8 rather than fixed post-hoc, since changing the bottleneck after observing per-fraud rank numbers would have violated the contract.

6. Statistical Defense

The frozen spec specifies four primary statistical analyses per cohort, all computed by scripts/06_validate.py. After observing the primary autoencoder result, I added one explicitly post-hoc baseline check.

Per-fraud rank within cohort with random-baseline comparison. All filings in a cohort are sorted by mean per-sentence reconstruction error descending (rank 1 = highest). Random-baseline expected rank for a cohort of size N is $(N+1)/2 \approx 6\text{-}7$. Hit@k for $k \in \{1, 3, 5\}$ is computed per cohort and aggregated.

Mann-Whitney U on per-sentence error distributions. For each cohort, the one-sided Mann-Whitney U statistic is computed on the per-sentence reconstruction-error distribution of the fraud filing's sentences vs. all peer sentences. The alternative is "fraud > peers." The test reports U, p-value, and rank-biserial correlation as effect size in [-1, 1]. Sentence counts feed directly into U, so the p-value scales with cohort sentence-count and must be read with the rank-biserial effect size, not in isolation.

Bootstrap and null permutation. I draw 1,000 filing-level bootstrap resamples within each cohort (peers resampled with replacement, fraud held fixed) and report the empirical 95% CI on the fraud's rank. I also run 1,000 within-cohort null-permutation iterations: under H0 (fraud and peers exchangeable) randomly assign one filing as the "fraud" and recompute the rank; the empirical p-value is the fraction of permutations under which the random fraud's rank is at least as extreme as observed.

Post-hoc TF-IDF + truncated-SVD baseline. A required baseline question for this work is whether a 1990s TF-IDF pipeline produces the same result. If yes, the autoencoder adds no evidence beyond the baseline on this dataset. To answer it, I implemented the exact analog of the autoencoder pipeline using TF-IDF features and truncated SVD as the bottleneck instead of MiniLM embeddings and a neural autoencoder. For each cohort I built the same training corpus and fit a TF-IDF vectorizer (English stop-word removal, sublinear term-frequency, 20,000-feature cap) followed by truncated SVD with 32 components. Per-sentence reconstruction error in the TF-IDF feature space is mean((x − UU^T x)^2) per sentence, aggregated to a filing-level mean. Outputs are at data/results/scores_tfidf.csv and data/results/per_fraud_metrics_tfidf.json.

Status of the post-hoc analyses. The TF-IDF baseline and the entity-masking ablation are post-hoc: both were added after observing the autoencoder's primary results, and the acceptance conditions for reporting them in the body (rather than the appendix) were authored at the same time. They are not confirmatory tests of a pre-registered hypothesis. Their role is diagnostic: they constrain the interpretation of the pre-registered null without changing it. Specifically, the TF-IDF baseline result says the autoencoder learned nothing that sentence-level vocabulary frequency did not already capture. The pre-registered primary result, including Enron's exclusion, is reported in Sections 7.1 and 7.2 unchanged whether or not these analyses had moved to the appendix.

7. Results

7.1 Methodological exclusion of Enron

Enron's FY2000 10-K was filed 2001-04-02. The earliest filing in any other cohort is Tyco's, dated 2001-12-28, 270 days later. Under the strict LOCO + time-controlled rule, Enron's eligible training set is empty: no autoencoder was trained, and no per-filing score, rank, Mann-Whitney result, or bootstrap CI can be reported for Enron without violating the frozen spec.

That outcome is correct under the contract, not a workaround for missing data. The pre-registered design was infeasible for Enron, and the design freeze did not catch the consequence. That is a finding about pre-registration, not fraud detection. It also illustrates a generalizable problem in held-out evaluation under temporal constraints: when held-out positive examples include the chronologically earliest one in the eligible universe, time-controlled training has no out-of-sample data to consume.

7.2 Per-fraud rank under the pre-registered autoencoder

Table 2. Pre-registered autoencoder, single-pass against frozen spec.

Fraud	N	Rank	Bootstrap 95% CI	MW p	Effect	Train sents
Lehman (LEH)	13	3	[1, 6]	1.3×10^{-9}	+0.12	3,367
HealthSouth (HRC)	12	7	[4, 10]	1.0	−0.17	2,283
Valeant (VRX)	13	8	[5, 11]	1.0	−0.19	4,327
Tyco (TYC)	13	9	[6, 12]	1.0	−0.14	861
WorldCom (WCOM)	13	12	[10, 13]	1.0	−0.15	1,477

Lehman is the only fraud ranked better than random expectation, and that result is n=1. The Mann-Whitney p-value of 1.3×10^{-9} on Lehman is small, but must be read alongside the effect size: rank-biserial of +0.12 (small) with 1,051 fraud sentences against 5,763 peer sentences. Sentence-level p-values scale aggressively with sentence count under fixed effect size; the rank-biserial effect is the safer summary statistic. The other four cohorts show negative effect sizes. Fraud per-sentence errors are statistically lower than peers', which means the autoencoder's signal on those cohorts is in the opposite of the predicted direction.

Across the five LOCO cohorts, aggregate hit@1 = 0.00, hit@3 = 0.20, hit@5 = 0.20, against random-baseline 0.08 / 0.23 / 0.39. The observed hit-rates fall at or below random expectation at every threshold. Hit@3 and hit@5 correspond entirely to Lehman.

7.3 Trivial baseline matches or exceeds the autoencoder on every cohort

Table 3. Post-hoc TF-IDF + SVD32 baseline vs autoencoder, same five cohorts.

Fraud	AE rank	TF-IDF rank	AE MW p	TF-IDF MW p	AE effect	TF-IDF effect
Lehman (LEH)	3	1	1.3×10^{-9}	5.0×10^{-22}	+0.12	+0.19
HealthSouth (HRC)	7	6	1.0	0.997	−0.17	−0.10
Valeant (VRX)	8	8	1.0	1.0	−0.19	−0.15
Tyco (TYC)	9	6	1.0	0.64	−0.14	−0.01
WorldCom (WCOM)	12	12	1.0	1.0	−0.15	−0.14

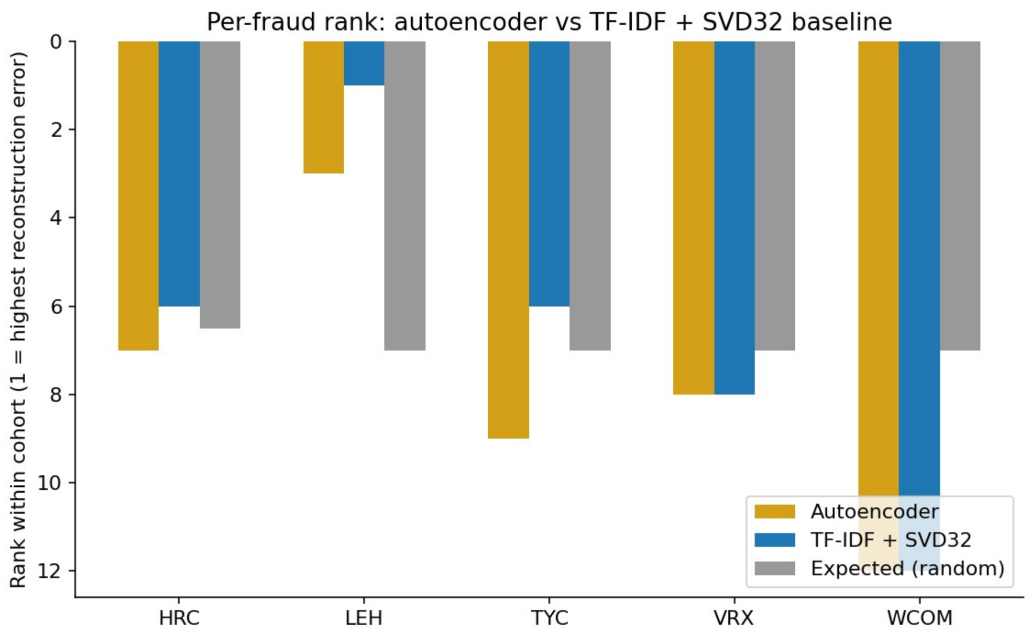

Figure 1. Per-fraud rank, pre-registered autoencoder vs post-hoc TF-IDF + SVD32 baseline. The trivial baseline matches the autoencoder on Valeant and WorldCom and outperforms it on Lehman, HealthSouth, and Tyco.

The TF-IDF + SVD32 baseline matches the autoencoder on Valeant and WorldCom and outperforms it on Lehman, HealthSouth, and Tyco. On Lehman, the trivial baseline lands the fraud at rank 1 of 13 with MW $p \approx 5 \times 10^{-22}$ and rank-biserial +0.19, both stronger than the autoencoder. Aggregate hit-rates for the trivial baseline are hit@1 = 0.20, hit@3 = 0.20, hit@5 = 0.40. The post-hoc TF-IDF baseline is the only analysis here with any aggregate hit-rate above random.

The interpretation requires care. The trivial baseline being competitive does not validate the autoencoder; it indicates that whatever signal exists at this dataset size is captured by sentence-level vocabulary distinctiveness alone. Whatever pattern appears here is text-distributional, not deep-learning-specific. At N = 6, on this signal, the 2020s neural model contributes nothing beyond a 1990s LSA baseline.

7.4 Lehman's positive result is unchanged by entity masking

The post-hoc entity-masking ablation tests whether MiniLM's web-pretraining seeded Lehman's rank by memorizing literal mentions of "Lehman," "Fuld," "Repo 105," or auditor names. For each cohort that has an autoencoder I hard-coded a per-cohort entity list (company name, key executives, auditor of record), replaced every case-insensitive whole-word match with [ENTITY] in the parsed MD&A, re-encoded, and rescored against the existing cohort autoencoder.

Table 4. Post-hoc entity-masking on the five LOCO cohorts.

Cohort	Replacements	Original rank	Masked rank	Δ
LEH	93	3 / 13	3 / 13	0
HRC	6	7 / 12	7 / 12	0
TYC	93	9 / 13	11 / 13	+2
VRX	31	8 / 13	9 / 13	+1
WCOM	267	12 / 13	12 / 13	0

Lehman's rank is unchanged at 3/13 after masking 93 mentions; exact-token name leakage does not explain this rank. The matching 93-replacement totals for LEH and TYC are coincidental: LEH = 64 "Lehman" + 29 "Lehman Brothers"; TYC = 87 "Tyco" + 6 "ADT".

Topical leakage (post-2008 commentary on repo accounting or off-balance-sheet vehicles) remains disclosed but unaddressed empirically.

7.5 Leave-one-fraud-out + interpretation

Removing each fraud in turn does not produce a regime under which the four-fraud aggregate beats random; removing Lehman drops hit@3 and hit@5 to 0.00. The aggregate result is entirely Lehman. Two readings are consistent with the data. Either the single signal has no out-of-sample power at $N = 6$, or MD&A language is too heterogeneous for a single autoencoder reconstruction-error signal to discriminate fraud from clean peers without supervision. Both yield the same conclusion: at this dataset size, on this signal, the answer to the driving question is no.

8. Conclusions, Limitations, Future Work

The results point to one negative finding and two qualifications. The pre-registered single-signal unsupervised novelty study on SEC 10-K MD&A text, evaluated under strict LOCO + time-controlled training against six historical fraud filings, produces a negative result. Of five evaluable cohorts, four rank fraud filings no better than the random-baseline expected rank ($(N+1)/2 \approx 7$); the single high-ranked cohort (Lehman, rank 3/13) is matched or exceeded by a 1990s-era TF-IDF + truncated-SVD trivial baseline. The first qualification is that the pre-registered design was infeasible for Enron under LOCO + time-controlled training because the chronologically earliest held-out positive has no eligible training data. That is a finding about pre-registration discipline, not fraud-detection capability, and a generalizable lesson: any held-out evaluation under temporal constraints must verify that every held-out positive admits at least one eligible training cohort under the proposed rule before the spec is frozen. The second qualification is that the trivial-baseline gap is the most decisive result. Without it, the project would have reported "Lehman is detectable" as the clean positive. With it, the honest claim is that Lehman is high-ranked under both methods, and the more expensive method does not beat the cheaper one. The lesson is that a result that loses to a baseline is not evidence for the neural model.

The work leaves two practical lessons. Pre-registration was load-bearing only because the frozen spec held under data that disappointed: I did not iterate the autoencoder, redefine "clean," or retreat from Enron's exclusion. The Enron exclusion is content rather than missing data because Section 7.1 reports it as a finding. The Lehman training-size confound, which is the obvious objection to the high-ranked cohort, is partially refuted by the project's own data: Valeant trained on more sentences (4,327) than Lehman (3,367) yet ranks 8/13, so training-corpus size alone does not explain Lehman's rank. The TF-IDF baseline reproduces the Lehman pattern using the same training corpora and same peer construction, which shows the pattern is not autoencoder-specific. It does not establish a fraud-specific Lehman signature; the result remains n=1 under both methods.

Limitations. N = 6 (effective N = 5) is too small for confidence intervals to constrain anything tightly. MiniLM pretraining contamination beyond literal-name leakage was not addressed empirically. SIC-2-digit + same-fiscal-year peer matching is coarse and admits residual heterogeneity. EDGAR's currently-reported SIC differs in some cases from the firm's at-time-of-filing SIC. The operational "clean" rule is editable-deny-list-only with partial coverage. The autoencoder bottleneck dimension was frozen at 32 without a sensitivity sweep.

Future work. The most informative single direction is change-detection within firm: instead of comparing a fraud filing's MD&A against industry peers, compare it against the same firm's prior-year MD&A. This controls for firm-specific style at the cost of losing the fraud-vs-clean comparison and converts the question from "is this filing anomalous against peers" to "is this filing anomalous against itself." Other directions: multi-section integration (Item 1A risk factors, Item 8 financial notes); structured-plus-unstructured fusion; a genuinely larger held-out set conditioned on AAER-confirmed fraud rather than the high-profile six; a supervised classical-ML baseline (logistic regression on TF-IDF features) once a larger fraud-label set is available. That would violate this study's contract but is the obvious follow-up.

References

Russell, S., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.

Burden, R. L., & Faires, J. D. (2010). Numerical Analysis (9th ed.). Brooks/Cole.

Cecchini, M., Aytug, H., Koehler, G. J., & Pathak, P. (2010). Making words work: Using financial text as a predictor of financial events. Decision Support Systems, 50(1), 164–175.

Hinton, G. E., & Salakhutdinov, R. R. (2006). Reducing the dimensionality of data with neural networks. Science, 313(5786), 504–507.

Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. The Journal of Finance, 66(1), 35–65.

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. Proc. EMNLP-IJCNLP 2019.

U.S. Securities and Exchange Commission. EDGAR full-text search and submissions JSON. https://www.sec.gov/edgar.

U.S. Securities and Exchange Commission. Accounting and Auditing Enforcement Releases. https://www.sec.gov/divisions/enforce/friactions.shtml.